

for all $a \in A$. For example, “=” and “ $\leq$ ” are reflexive relations on $\mathbb{N}$, but “ $<$ ” is not. The relation R is *symmetric* if

$a R b$ implies $b R a$

for all $a, b \in A$. For example, “=” is symmetric, but “ $<$ ” and “ $\leq$ ” are not. The relation R is *transitive* if

$a R b$ and $b R c$ imply $a R c$

for all $a, b, c \in A$. For example, the relations “ $<$,” “ $\leq$,” and “=” are transitive, but the relation $R = \{(a, b) : a, b \in \mathbb{N} \text{ and } a = b - 1\}$ is not, since $3 R 4$ and $4 R 5$ do not imply $3 R 5$.

A relation that is reflexive, symmetric, and transitive is an *equivalence relation*. For example, “=” is an equivalence relation on the natural numbers, but “ $<$ ” is not. If R is an equivalence relation on a set A , then for $a \in A$, the *equivalence class* of a is the set $[a] = \{b \in A : a R b\}$, that is, the set of all elements equivalent to a . For example, if we define $R = \{(a, b) : a, b \in \mathbb{N} \text{ and } a + b \text{ is an even number}\}$, then R is an equivalence relation, since $a + a$ is even (reflexive), $a + b$ is even implies $b + a$ is even (symmetric), and $a + b$ is even and $b + c$ is even imply $a + c$ is even (transitive). The equivalence class of 4 is $[4] = \{0, 2, 4, 6, \dots\}$, and the equivalence class of 3 is $[3] = \{1, 3, 5, 7, \dots\}$. A basic theorem of equivalence classes is the following.

Theorem B.1 (An equivalence relation is the same as a partition)

The equivalence classes of any equivalence relation R on a set A form a partition of A , and any partition of A determines an equivalence relation on A for which the sets in the partition are the equivalence classes.

Proof For the first part of the proof, we must show that the equivalence classes of R are nonempty, pairwise-disjoint sets whose union is A . Because R is reflexive, $a \in [a]$, and so the equivalence classes are nonempty. Moreover, since every element $a \in A$ belongs to the equivalence class $[a]$, the union of the equivalence classes is A . It remains to show that the equivalence classes are pairwise disjoint, that is, if two equivalence classes $[a]$ and $[b]$ have an element c in common, then they are in fact the same set. Suppose that $a R c$ and $b R c$. Symmetry gives that $c R b$ and, by transitivity, $a R b$. Thus, we have $x R a$ for any arbitrary element $x \in [a]$ and, by transitivity, $x R b$, and thus $[a] \subseteq [b]$. Similarly, $[b] \subseteq [a]$, and thus $[a] = [b]$.

For the second part of the proof, let $\mathcal{A} = \{A_i\}$ be a partition of A , and define $R = \{(a, b) : \text{there exists } i \text{ such that } a \in A_i \text{ and } b \in A_i\}$. We claim that R is an equivalence relation on A . Reflexivity holds, since $a \in A_i$ implies $a R a$. Symmetry holds, because if $a R b$, then a and b belong to the same set A_i , and

hence $b R a$. If $a R b$ and $b R c$, then all three elements are in the same set A_i , and thus $a R c$ and transitivity holds. To see that the sets in the partition are the equivalence classes of R , observe that if $a \in A_i$, then $x \in [a]$ implies $x \in A_i$, and $x \in A_i$ implies $x \in [a]$. ■

A binary relation R on a set A is *antisymmetric* if
 $a R b$ and $b R a$ imply $a = b$.

For example, the “ $\leq$ ” relation on the natural numbers is antisymmetric, since $a \leq b$ and $b \leq a$ imply $a = b$. A relation that is reflexive, antisymmetric, and transitive is a *partial order*, and we call a set on which a partial order is defined a *partially ordered set*. For example, the relation “is a descendant of” is a partial order on the set of all people (if we view individuals as being their own descendants).

In a partially ordered set A , there may be no single “maximum” element a such that $b R a$ for all $b \in A$. Instead, the set may contain several *maximal* elements a such that for no $b \in A$, where $b \neq a$, is it the case that $a R b$. For example, a collection of different-sized boxes may contain several maximal boxes that don’t fit inside any other box, yet it has no single “maximum” box into which any other box will fit.³

A relation R on a set A is a *total relation* if for all $a, b \in A$, we have $a R b$ or $b R a$ (or both), that is, if every pairing of elements of A is related by R . A partial order that is also a total relation is a *total order* or *linear order*. For example, the relation “ $\leq$ ” is a total order on the natural numbers, but the “is a descendant of” relation is not a total order on the set of all people, since there are individuals neither of whom is descended from the other. A total relation that is transitive, but not necessarily either symmetric or antisymmetric, is a *total preorder*.

Exercises

B.2-1

Prove that the subset relation “ $\subseteq$ ” on all subsets of $\mathbb{Z}$ is a partial order but not a total order.

B.2-2

Show that for any positive integer n , the relation “equivalent modulo n ” is an equivalence relation on the integers. (We say that $a \equiv b \pmod{n}$ if there exists an integer q such that $a - b = qn$.) Into what equivalence classes does this relation partition the integers?

³ To be precise, in order for the “fit inside” relation to be a partial order, we need to view a box as fitting inside itself.

B.2-3

Give examples of relations that are

- a.* reflexive and symmetric but not transitive,
- b.* reflexive and transitive but not symmetric,
- c.* symmetric and transitive but not reflexive.

B.2-4

Let S be a finite set, and let R be an equivalence relation on $S \times S$. Show that if in addition R is antisymmetric, then the equivalence classes of S with respect to R are singletons.

B.2-5

Professor Narcissus claims that if a relation R is symmetric and transitive, then it is also reflexive. He offers the following proof. By symmetry, $a R b$ implies $b R a$. Transitivity, therefore, implies $a R a$. Is the professor correct?

B.3 Functions

Given two sets A and B , a *function* f is a binary relation on A and B such that for all $a \in A$, there exists precisely one $b \in B$ such that $(a, b) \in f$. The set A is called the *domain* of f , and the set B is called the *codomain* of f . We sometimes write $f : A \rightarrow B$, and if $(a, b) \in f$, we write $b = f(a)$, since the choice of a uniquely determines b .

Intuitively, the function f assigns an element of B to each element of A . No element of A is assigned two different elements of B , but the same element of B can be assigned to two different elements of A . For example, the binary relation

$$f = \{(a, b) : a, b \in \mathbb{N} \text{ and } b = a \bmod 2\}$$

is a function $f : \mathbb{N} \rightarrow \{0, 1\}$, since for each natural number a , there is exactly one value b in $\{0, 1\}$ such that $b = a \bmod 2$. For this example, $0 = f(0)$, $1 = f(1)$, $0 = f(2)$, $1 = f(3)$, etc. In contrast, the binary relation

$$g = \{(a, b) : a, b \in \mathbb{N} \text{ and } a + b \text{ is even}\}$$

is not a function, since $(1, 3)$ and $(1, 5)$ are both in g , and thus for the choice $a = 1$, there is not precisely one b such that $(a, b) \in g$.

Given a function $f : A \rightarrow B$, if $b = f(a)$, we say that a is the *argument* of f and that b is the *value* of f at a . We can define a function by stating its value for

every element of its domain. For example, we might define $f(n) = 2n$ for $n \in \mathbb{N}$, which means $f = \{(n, 2n) : n \in \mathbb{N}\}$. Two functions f and g are **equal** if they have the same domain and codomain and if $f(a) = g(a)$ for all a in the domain.

A **finite sequence** of length n is a function f whose domain is the set of n integers $\{0, 1, \dots, n-1\}$. We often denote a finite sequence by listing its values in angle brackets: $\langle f(0), f(1), \dots, f(n-1) \rangle$. An **infinite sequence** is a function whose domain is the set $\mathbb{N}$ of natural numbers. For example, the Fibonacci sequence, defined by recurrence (3.31), is the infinite sequence $\langle 0, 1, 1, 2, 3, 5, 8, 13, 21, \dots \rangle$.

When the domain of a function f is a Cartesian product, we often omit the extra parentheses surrounding the argument of f . For example, if we have a function $f : A_1 \times A_2 \times \dots \times A_n \rightarrow B$, we write $b = f(a_1, a_2, \dots, a_n)$ instead of writing $b = f((a_1, a_2, \dots, a_n))$. We also call each a_i an **argument** to the function f , though technically f has just a single argument, which is the n -tuple $(a_1, a_2, \dots, a_n)$.

If $f : A \rightarrow B$ is a function and $b = f(a)$, then we sometimes say that b is the **image** of a under f . The image of a set $A' \subseteq A$ under f is defined by

$$f(A') = \{b \in B : b = f(a) \text{ for some } a \in A'\}.$$

The **range** of f is the image of its domain, that is, $f(A)$. For example, the range of the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(n) = 2n$ is $f(\mathbb{N}) = \{m : m = 2n \text{ for some } n \in \mathbb{N}\}$, in other words, the set of nonnegative even integers.

A function is a **surjection** if its range is its codomain. For example, the function $f(n) = \lfloor n/2 \rfloor$ is a surjective function from $\mathbb{N}$ to $\mathbb{N}$, since every element in $\mathbb{N}$ appears as the value of f for some argument. In contrast, the function $f(n) = 2n$ is not a surjective function from $\mathbb{N}$ to $\mathbb{N}$, since no argument to f can produce any odd natural number as a value. The function $f(n) = 2n$ is, however, a surjective function from the natural numbers to the even numbers. A surjection $f : A \rightarrow B$ is sometimes described as mapping A **onto** B . When we say that f is onto, we mean that it is surjective.

A function $f : A \rightarrow B$ is an **injection** if distinct arguments to f produce distinct values, that is, if $a \neq a'$ implies $f(a) \neq f(a')$. For example, the function $f(n) = 2n$ is an injective function from $\mathbb{N}$ to $\mathbb{N}$, since each even number b is the image under f of at most one element of the domain, namely $b/2$. The function $f(n) = \lfloor n/2 \rfloor$ is not injective, since the value 1 is produced by two arguments: $f(2) = 1$ and $f(3) = 1$. An injection is sometimes called a **one-to-one** function.

A function $f : A \rightarrow B$ is a **bijection** if it is injective and surjective. For example, the function $f(n) = (-1)^n \lfloor n/2 \rfloor$ is a bijection from $\mathbb{N}$ to $\mathbb{Z}$:

$$\begin{aligned}
0 &\rightarrow 0, \\
1 &\rightarrow -1, \\
2 &\rightarrow 1, \\
3 &\rightarrow -2, \\
4 &\rightarrow 2, \\
&\vdots
\end{aligned}$$

The function is injective, since no element of $\mathbb{Z}$ is the image of more than one element of $\mathbb{N}$. It is surjective, since every element of $\mathbb{Z}$ appears as the image of some element of $\mathbb{N}$. Hence, the function is bijective. A bijection is sometimes called a *one-to-one correspondence*, since it pairs elements in the domain and codomain. A bijection from a set A to itself is sometimes called a *permutation*.

When a function f is bijective, we define its *inverse* f^{-1} as

$$f^{-1}(b) = a \text{ if and only if } f(a) = b.$$

For example, the inverse of the function $f(n) = (-1)^n \lceil n/2 \rceil$ is

$$f^{-1}(m) = \begin{cases} 2m & \text{if } m \geq 0, \\ -2m - 1 & \text{if } m < 0. \end{cases}$$

Exercises

B.3-1

Let A and B be finite sets, and let $f : A \rightarrow B$ be a function. Show the following:

- a. If f is injective, then $|A| \leq |B|$.
- b. If f is surjective, then $|A| \geq |B|$.

B.3-2

Is the function $f(x) = x + 1$ bijective when the domain and the codomain are the set $\mathbb{N}$? Is it bijective when the domain and the codomain are the set $\mathbb{Z}$?

B.3-3

Give a natural definition for the inverse of a binary relation such that if a relation is in fact a bijective function, its relational inverse is its functional inverse.

★ B.3-4

Give a bijection from $\mathbb{Z}$ to $\mathbb{Z} \times \mathbb{Z}$.

B.4 Graphs

This section presents two kinds of graphs: directed and undirected. Certain definitions in the literature differ from those given here, but for the most part, the differences are slight. Section 20.1 shows how to represent graphs in computer memory.

A **directed graph** (or **digraph**) G is a pair (V, E) , where V is a finite set and E is a binary relation on V . The set V is called the **vertex set** of G , and its elements are called **vertices** (singular: **vertex**). The set E is called the **edge set** of G , and its elements are called **edges**. Figure B.2(a) is a pictorial representation of a directed graph on the vertex set $\{1, 2, 3, 4, 5, 6\}$. Vertices are represented by circles in the figure, and edges are represented by arrows. **Self-loops**—edges from a vertex to itself—are possible.

In an **undirected graph** $G = (V, E)$, the edge set E consists of *unordered* pairs of vertices, rather than ordered pairs. That is, an edge is a set $\{u, v\}$, where $u, v \in V$ and $u \neq v$. By convention, we use the notation (u, v) for an edge, rather than the set notation $\{u, v\}$, and we consider (u, v) and (v, u) to be the same edge. In an undirected graph, self-loops are forbidden, so that every edge consists of two distinct vertices. Figure B.2(b) shows an undirected graph on the vertex set $\{1, 2, 3, 4, 5, 6\}$.

Many definitions for directed and undirected graphs are the same, although certain terms have slightly different meanings in the two contexts. If (u, v) is an edge in a directed graph $G = (V, E)$, we say that (u, v) is **incident from** or **leaves** vertex u and is **incident to** or **enters** vertex v . For example, the edges leaving vertex 2

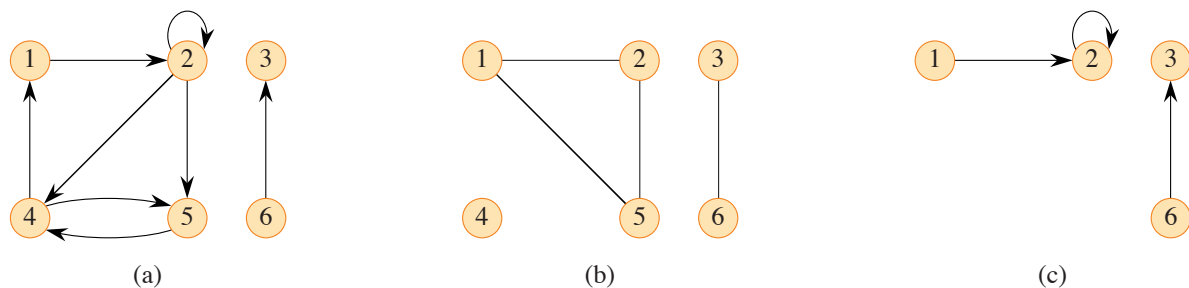


Figure B.2 Directed and undirected graphs. (a) A directed graph $G = (V, E)$, where $V = \{1, 2, 3, 4, 5, 6\}$ and $E = \{(1, 2), (2, 2), (2, 4), (2, 5), (4, 1), (4, 5), (5, 4), (6, 3)\}$. The edge $(2, 2)$ is a self-loop. (b) An undirected graph $G = (V, E)$, where $V = \{1, 2, 3, 4, 5, 6\}$ and $E = \{(1, 2), (1, 5), (2, 5), (3, 6)\}$. The vertex 4 is isolated. (c) The subgraph of the graph in part (a) induced by the vertex set $\{1, 2, 3, 6\}$.

in Figure B.2(a) are $(2, 2)$, $(2, 4)$, and $(2, 5)$. The edges entering vertex 2 are $(1, 2)$ and $(2, 2)$. If (u, v) is an edge in an undirected graph $G = (V, E)$, we say that (u, v) is **incident on** vertices u and v . In Figure B.2(b), the edges incident on vertex 2 are $(1, 2)$ and $(2, 5)$.

If (u, v) is an edge in a graph $G = (V, E)$, we say that vertex v is **adjacent** to vertex u . When the graph is undirected, the adjacency relation is symmetric. When the graph is directed, the adjacency relation is not necessarily symmetric. If v is adjacent to u in a directed graph, we can write $u \rightarrow v$. In parts (a) and (b) of Figure B.2, vertex 2 is adjacent to vertex 1, since the edge $(1, 2)$ belongs to both graphs. Vertex 1 is *not* adjacent to vertex 2 in Figure B.2(a), since the edge $(2, 1)$ is absent.

The **degree** of a vertex in an undirected graph is the number of edges incident on it. For example, vertex 2 in Figure B.2(b) has degree 2. A vertex whose degree is 0, such as vertex 4 in Figure B.2(b), is **isolated**. In a directed graph, the **out-degree** of a vertex is the number of edges leaving it, and the **in-degree** of a vertex is the number of edges entering it. The **degree** of a vertex in a directed graph is its in-degree plus its out-degree. Vertex 2 in Figure B.2(a) has in-degree 2, out-degree 3, and degree 5.

A **path of length** k from a vertex u to a vertex u' in a graph $G = (V, E)$ is a sequence $\langle v_0, v_1, v_2, \dots, v_k \rangle$ of vertices such that $u = v_0$, $u' = v_k$, and $(v_{i-1}, v_i) \in E$ for $i = 1, 2, \dots, k$. The length of the path is the number of edges in the path, which is 1 less than the number of vertices in the path. The path **contains** the vertices $v_0, v_1, \dots, v_k$ and the edges $(v_0, v_1), (v_1, v_2), \dots, (v_{k-1}, v_k)$. (There is always a 0-length path from u to u .) If there is a path p from u to u' , we say that u' is **reachable** from u via p , which we can write as $u \xrightarrow{p} u'$. A path is **simple**⁴ if all vertices in the path are distinct. In Figure B.2(a), the path $\langle 1, 2, 5, 4 \rangle$ is a simple path of length 3. The path $\langle 2, 5, 4, 5 \rangle$ is not simple. A **subpath** of path $p = \langle v_0, v_1, \dots, v_k \rangle$ is a contiguous subsequence of its vertices. That is, for any $0 \leq i \leq j \leq k$, the subsequence of vertices $\langle v_i, v_{i+1}, \dots, v_j \rangle$ is a subpath of p .

In a directed graph, a path $\langle v_0, v_1, \dots, v_k \rangle$ forms a **cycle** if $v_0 = v_k$ and the path contains at least one edge. The cycle is **simple** if, in addition, $v_1, v_2, \dots, v_k$ are distinct. A cycle consisting of k vertices has **length** k . A self-loop is a cycle of length 1. Two paths $\langle v_0, v_1, v_2, \dots, v_{k-1}, v_0 \rangle$ and $\langle v'_0, v'_1, v'_2, \dots, v'_{k-1}, v'_0 \rangle$ form the same cycle if there exists an integer j such that $v'_i = v_{(i+j) \bmod k}$ for $i = 0, 1, \dots, k-1$. In Figure B.2(a), the path $\langle 1, 2, 4, 1 \rangle$ forms the same cycle as the paths $\langle 2, 4, 1, 2 \rangle$ and $\langle 4, 1, 2, 4 \rangle$. This cycle is simple, but the cycle $\langle 1, 2, 4, 5, 4, 1 \rangle$ is not. The cycle $\langle 2, 2 \rangle$ formed by the edge $(2, 2)$ is a self-loop. A directed graph

⁴ Some authors refer to what we call a path as a “walk” and to what we call a simple path as just a “path.”

with no self-loops is **simple**. In an undirected graph, a path $\langle v_0, v_1, \dots, v_k \rangle$ forms a **cycle** if $k > 0$, $v_0 = v_k$, and all edges on the path are distinct. The cycle is **simple** if $v_1, v_2, \dots, v_k$ are distinct. For example, in Figure B.2(b), the path $\langle 1, 2, 5, 1 \rangle$ is a simple cycle. A graph with no simple cycles is **acyclic**.

An undirected graph is **connected** if every vertex is reachable from all other vertices. The **connected components** of an undirected graph are the equivalence classes of vertices under the “is reachable from” relation. The graph shown in Figure B.2(b) has three connected components: $\{1, 2, 5\}$, $\{3, 6\}$, and $\{4\}$. Every vertex in the connected component $\{1, 2, 5\}$ is reachable from every other vertex in $\{1, 2, 5\}$. An undirected graph is connected if it has exactly one connected component. The edges of a connected component are those that are incident on only the vertices of the component. In other words, edge (u, v) is an edge of a connected component only if both u and v are vertices of the component.

A directed graph is **strongly connected** if every two vertices are reachable from each other. The **strongly connected components** of a directed graph are the equivalence classes of vertices under the “are mutually reachable” relation. A directed graph is strongly connected if it has only one strongly connected component. The graph in Figure B.2(a) has three strongly connected components: $\{1, 2, 4, 5\}$, $\{3\}$, and $\{6\}$. All pairs of vertices in $\{1, 2, 4, 5\}$ are mutually reachable. The vertices $\{3, 6\}$ do not form a strongly connected component, since vertex 6 cannot be reached from vertex 3.

Two graphs $G = (V, E)$ and $G' = (V', E')$ are **isomorphic** if there exists a bijection $f : V \rightarrow V'$ such that $(u, v) \in E$ if and only if $(f(u), f(v)) \in E'$. In other words, G and G' are isomorphic if the vertices of G can be relabeled to be vertices of G' , maintaining the corresponding edges in G and G' . Figure B.3(a) shows a pair of isomorphic graphs G and G' with respective vertex sets $V = \{1, 2, 3, 4, 5, 6\}$ and $V' = \{u, v, w, x, y, z\}$. The mapping from V to V' given by $f(1) = u, f(2) = v, f(3) = w, f(4) = x, f(5) = y, f(6) = z$ provides the required bijective function. The graphs in Figure B.3(b) are not isomorphic. Although both graphs have 5 vertices and 7 edges, the top graph has a vertex of degree 4 and the bottom graph does not.

We say that a graph $G' = (V', E')$ is a **subgraph** of $G = (V, E)$ if $V' \subseteq V$ and $E' \subseteq E$. Given a set $V' \subseteq V$, the subgraph of G **induced** by V' is the graph $G' = (V', E')$, where

$$E' = \{(u, v) \in E : u, v \in V'\} .$$

The subgraph induced by the vertex set $\{1, 2, 3, 6\}$ in Figure B.2(a) appears in Figure B.2(c) and has the edge set $\{(1, 2), (2, 2), (6, 3)\}$.

Given an undirected graph $G = (V, E)$, the **directed version** of G is the directed graph $G' = (V, E')$, where $(u, v) \in E'$ if and only if $(u, v) \in E$. That is, each undirected edge (u, v) in G turns into two directed edges, (u, v) and (v, u) , in the

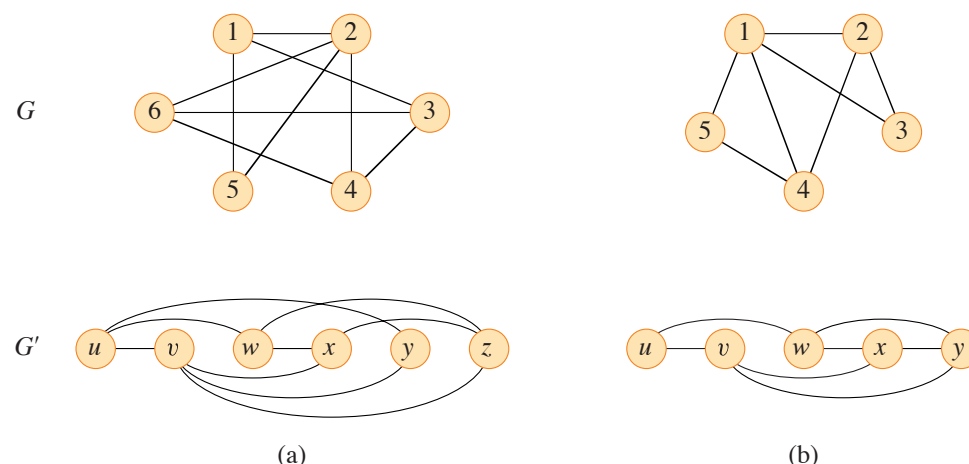


Figure B.3 (a) A pair of isomorphic graphs. The vertices of the top graph are mapped to the vertices of the bottom graph by $f(1) = u, f(2) = v, f(3) = w, f(4) = x, f(5) = y, f(6) = z$. (b) Two graphs that are not isomorphic. The top graph has a vertex of degree 4, and the bottom graph does not.

directed version. Given a directed graph $G = (V, E)$, the **undirected version** of G is the undirected graph $G' = (V, E')$, where $(u, v) \in E'$ if and only if $u \neq v$ and E contains at least one of the edges (u, v) and (v, u) . That is, the undirected version contains the edges of G “with their directions removed” and with self-loops eliminated. (Since (u, v) and (v, u) are the same edge in an undirected graph, the undirected version of a directed graph contains it only once, even if the directed graph contains both edges (u, v) and (v, u) .) In a directed graph $G = (V, E)$, a **neighbor** of a vertex u is any vertex that is adjacent to u in the undirected version of G . That is, v is a neighbor of u if $u \neq v$ and either $(u, v) \in E$ or $(v, u) \in E$. In an undirected graph, u and v are neighbors if they are adjacent.

Several kinds of graphs have special names. A **complete graph** is an undirected graph in which every pair of vertices is adjacent. An undirected graph $G = (V, E)$ is **bipartite** if V can be partitioned into two sets V_1 and V_2 such that $(u, v) \in E$ implies either $u \in V_1$ and $v \in V_2$ or $u \in V_2$ and $v \in V_1$. That is, all edges go between the two sets V_1 and V_2 . An acyclic, undirected graph is a **forest**, and a connected, acyclic, undirected graph is a **(free) tree** (see Section B.5). We often take the first letters of “directed acyclic graph” and call such a graph a **dag**.

There are two variants of graphs that you may occasionally encounter. A **multi-graph** is like an undirected graph, but it can have both multiple edges between vertices (such as two distinct edges (u, v) and (u, v)) and self-loops. A **hypergraph** is like an undirected graph, but each **hyperedge**, rather than connecting two vertices,

connects an arbitrary subset of vertices. Many algorithms written for ordinary directed and undirected graphs can be adapted to run on these graphlike structures.

The **contraction** of an undirected graph $G = (V, E)$ by an edge $e = (u, v)$ is a graph $G' = (V', E')$, where $V' = V - \{u, v\} \cup \{x\}$ and x is a new vertex. The set of edges E' is formed from E by deleting the edge (u, v) and, for each vertex w adjacent to u or v , deleting whichever of (u, w) and (v, w) belongs to E and adding the new edge (x, w) . In effect, u and v are “contracted” into a single vertex.

Exercises

B.4-1

Attendees of a faculty party shake hands to greet each other, with every pair of professors shaking hands one time. Each professor remembers the number of times he or she shook hands. At the end of the party, the department head asks the professors for their totals and adds them all up. Show that the result is even by proving the **handshaking lemma**: if $G = (V, E)$ is an undirected graph, then

$$\sum_{v \in V} \text{degree}(v) = 2|E|.$$

B.4-2

Show that if a directed or undirected graph contains a path between two vertices u and v , then it contains a simple path between u and v . Show that if a directed graph contains a cycle, then it contains a simple cycle.

B.4-3

Show that any connected, undirected graph $G = (V, E)$ satisfies $|E| \geq |V| - 1$.

B.4-4

Verify that in an undirected graph, the “is reachable from” relation is an equivalence relation on the vertices of the graph. Which of the three properties of an equivalence relation hold in general for the “is reachable from” relation on the vertices of a directed graph?

B.4-5

What is the undirected version of the directed graph in Figure B.2(a)? What is the directed version of the undirected graph in Figure B.2(b)?

B.4-6

Show how a bipartite graph can represent a hypergraph by letting incidence in the hypergraph correspond to adjacency in the bipartite graph. (*Hint*: Let one set of